

ALARI ILVES

Senior AI/ML Engineer | LLM, RAG, NLP | Python, PyTorch, AWS | FastAPI

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SUMMARY

Senior **AI/ML Software Engineer** with 10+ years of experience building and operating production-grade intelligent systems across **FinTech, TravelTech, EdTech, Developer Platforms**, and **enterprise SaaS environments**. Specialized in applying **LLMs, NLP, RAG**, and **machine learning pipelines** to real-world problems, with a strong focus on reliability, latency, and data quality in production systems. Experienced in delivering end-to-end solutions—from data ingestion and model development to **scalable cloud deployment, microservices architecture**, and **observability dashboards** — enabling practical, explainable, and data-driven decision-making at scale.

PROFESSIONAL EXPERIENCE

Senior AI / ML Software Engineer

Yalantis | Warszawa, Poland | Nov 2021 – Apr 2026

- Led the backend and ML architecture for a **voice-driven concierge platform** used by hotel operators, handling real guest requests (booking updates, service orders, FAQs) across multiple regions; contributed to automation that enabled ~75% of guest interactions to be handled without manual intervention.
- Built and continuously improved **multilingual NLP pipelines** using **Python, PyTorch, Hugging Face, and spaCy**, tuning intent classification and entity extraction against real production logs rather than static datasets.
- Introduced **LLM-based response generation (OpenAI API)** in a controlled way—kept business logic deterministic and used LLMs only where they added value, with guardrails and fallback flows to prevent hallucinations in customer-facing responses.
- Designed a **hybrid retrieval layer (Elasticsearch + vector search via FAISS/Pinecone)** to serve hotel policies and guest context; spent significant effort on ranking quality, cache strategy, and keeping latency stable under load.
- Built a **distributed microservices setup** using **FastAPI and gRPC**, with **Celery, Redis, and Kafka** for async workflows (speech processing, enrichment, integrations); tuned queues and backpressure to handle traffic spikes without cascading failures.
- Developed **data pipelines (Airflow, Spark, PostgreSQL)** to process booking logs and interaction data into training datasets; worked closely with real data issues (missing fields, noisy inputs, schema drift).
- Owned parts of the **AWS deployment (EC2, S3, Lambda, RDS, SageMaker)** using **Docker/Kubernetes + Terraform**, with emphasis on safe rollouts, observability, and rollback rather than over-engineering infra.
- Integrated with external hotel systems (**PMS/CRM**) via **REST, GraphQL, and WebSockets**, handling unreliable APIs, rate limits, and inconsistent contracts.
- Built **monitoring and dashboards (Prometheus, Grafana, React)** to track latency, error rates, and system behavior; these insights helped reduce operational overhead and contributed to ~40% reduction in manual workflows and support costs.

EDUCATION

Bachelor's Degree

in Computer Science

University of Tartu Narva College

Sep 2010 - Jun 2013

COMPETENCIES

- LLM & Generative AI Systems
- Machine Learning & NLP
- Data Pipelines & Feature Engineering
- Search & Retrieval Systems
- Backend & Distributed Systems
- Cloud & MLOps
- System Design & Scalability
- Observability & Analytics

SKILLS

AI / ML

Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, NLP (spaCy, NLTK), Transformers, Hugging Face LLM, RAG, Prompt Engineering, OpenAI, GitHub Copilot, Claude, Evaluation & Guardrails

AI / ML Software Engineer

Eremenko & Polomani | Tallinn, Estonia | May 2018 – Oct 2021

- Built a system to process incoming booking emails and extract structured data (dates, locations, preferences), reducing manual parsing work for operations teams.
- Implemented NLP pipelines with **spaCy**, **NLTK**, and **Transformers**, focusing on robustness rather than perfect accuracy due to messy, real-world email formats.
- Trained and evaluated models using **Scikit-learn**, **XGBoost**, and **LightGBM** for tasks like classification and ranking; kept models simple and explainable since outputs affected operational decisions.
- Developed backend services with **FastAPI**, using **Celery** + **RabbitMQ** + **Redis** to handle asynchronous email processing and retries.
- Built preprocessing pipelines with **Pandas** and **Airflow**, cleaning HTML-heavy email data and converting it into usable training datasets.
- Deployed workloads on **Azure (AKS, Azure ML, Functions)**; spent a good amount of time debugging deployment issues and handling scaling inconsistencies.
- Created lightweight dashboards (mainly **Power BI**) so non-technical users could track processing results and error cases.

AI / ML Software Engineer

Terratra | Maardu, Estonia | Sep 2015 – Apr 2018

- Worked on an automated grading system for educational platforms, helping reduce manual grading load while keeping results interpretable for teachers.
- Built evaluation pipelines combining rule-based logic with **ML models (PyTorch, Transformers)** to handle essays, structured answers, and simple code submissions.
- Introduced retrieval-based approaches (early **RAG-style patterns**) to reference grading rubrics and course materials instead of relying purely on model outputs.
- Designed backend workflows using **Kafka**, **Redis**, and **PostgreSQL** to process large batches of submissions asynchronously.
- Deployed services on **GCP (GKE, Cloud Run)** using **Docker/Kubernetes**, focusing on cost control and predictable scaling during exam periods.
- Integrated with LMS platforms via **REST APIs** and **OAuth**, dealing with version mismatches and inconsistent payloads.

Machine Learning Engineer

Golems GABB | Tallinn, Estonia | Jul 2013 – Aug 2015

- Contributed to an internal ML-driven platform for developer workflow automation, focusing on code understanding and task prediction from large codebases.
- Built code analysis pipelines using AST parsing and structural feature extraction (dependencies, call graphs, file relationships) to generate ML-ready datasets.
- Developed feature engineering workflows with **Python**, **NumPy**, and **Pandas**, transforming repository metadata into structured inputs for downstream models.
- Implemented classification and ranking models (**Scikit-learn**) to recommend relevant files and implementation patterns based on historical commit data.
- Applied early embedding techniques (**TF-IDF**, **word2vec-style**) to model code similarity and improve search and recommendation relevance.
- Designed offline evaluation pipelines using historical data to validate model performance and reduce noise in predictions.

Data Engineering

Pandas, NumPy, Apache Airflow, Apache Spark, ETL Pipelines, Feature Engineering, Data Preprocessing

Search & Retrieval

Elasticsearch, Vector Databases: FAISS, Pinecone, Semantic Search, Hybrid Retrieval Systems

Backend & Distributed Systems

FastAPI, Django, Node.js, gRPC, REST APIs, GraphQL, WebSockets, Celery, Kafka, RabbitMQ, Redis, Microservices Architecture, Async Processing

Databases & Storage

PostgreSQL, MySQL, MongoDB, SQL, Data Modeling, Data Lakes

Cloud & MLOps

AWS (EC2, S3, Lambda, SageMaker, RDS, CloakWatch,), Azure (AKS, Azure ML, Functions), GCP (GKE, Vertex AI, Cloud Run), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Azure DevOps), Model Deployment, Scalable Inference

Monitoring & Observability

Prometheus, Grafana, Logging, Metrics, Alerting, OpenTelemetry, ELK stack (Elasticsearch, Logstash, Kibana), SLA/SLO monitoring and incident response

Visualization & Dashboards

Power BI, React, D3.js, Real-time Analytics Dashboards, Data visualization (Plotly and Chart.js), KPI tracking and business metrics visualization

PROJECTS

AI Voice Concierge for Hotels and Travel Operators

<https://delight.ai>

An AI-powered platform was implemented to streamline hotel booking workflows and guest service operations in the travel and hospitality industry. The system combines machine learning, large language models, conversational AI, and predictive analytics to automate guest communication, process booking inquiries, and manage concierge services. Requests from email, voice, phone, or mobile channels are automatically analyzed to detect language, classify intent, and extract key booking details. The platform then checks availability, identifies events or peak periods, and recommends optimal hotel allocations or service actions while generating response suggestions for staff. Integrated with core hospitality systems—including PMS, reservation platforms, POS, CRM, ERP, and smart-room technologies—it creates a unified operational layer that enables real-time coordination across departments. By leveraging guest profiles and historical data for personalization and demand forecasting, the solution reduces response times from minutes to seconds, supports multilingual interactions at scale, and significantly improves operational efficiency and guest experience.

AI-Driven Automated Grading and Feedback Platform for Modern Education

<https://www.braintrust.dev>

An AI-powered grading system was developed to automate and standardize student assessment in digital learning environments. Using large language models and Retrieval-Augmented Generation (RAG), the platform evaluates assignments against structured rubrics and generates clear, explainable feedback. It supports multiple submission types—including essays, code, math, and audio—while referencing indexed course materials, rubrics, and policies to justify scores and ensure grading consistency. Integrated with Learning Management Systems (LMS), the system automates grading, feedback delivery, and regrade workflows. A human-in-the-loop review process handles low-confidence cases to maintain fairness and accuracy. Built on scalable and secure infrastructure, the platform processes large volumes of submissions efficiently, reducing manual grading workload while improving feedback quality, transparency, and learning outcomes.